Chapter 1

Beginning Combinatorics

Discrete probability problems often include some counting. For example, we figured out that there were 36 different ways the two dice could land, but all of them summed to some number 2 through 12. How many different ways could three eight-sided dice come up? We would need to count them, right? As the numbers get bigger, we will need some tricks so that we don't need to write them all down and count them one-by-one.

The branch of mathematics that focuses on tricks for counting is called *combinatorics*.

How can we be sure that there were 36 different configurations for the two 6-sided dice? The first die could have come up as any one of six numbers. For each of those, the second could have come up with any one of six numbers. Thus, the number of possibilities is $6 \times 6 = 36$.

How many different configurations for three 8-sided dice? $8 \times 8 \times 8 = 8^3 = 512$.

What about seven dice, each with 20 sides? There would be $20^7 = 1,280,000,000$ configurations. See, aren't you glad we don't need to write them all down?

Now, let's say that six people (Anne, Brock, Carl, Dev, Edgar, and Fred) are going to run a race. You have to make a plaque that says who won first place, who won second place, and who won third. If you want to get all the possible plaques created beforehand, and just pull the right one out as soon as the race ends, how many plaques would you need to get engraved?

Since you're making these plaques before the race even happens, you need one plaque for every possible outcome. So think of it as filling in three blanks: first place, second place, and third place. Any of the 6 people could fill the first blank. Whoever fills that blank can't also fill the second blank — a person can't come in both first *and* second — so only 5 people are left to fill the second blank. Likewise, only 4 people remain to fill the third blank, since two people have already been assigned to the first two spots. Multiplying these together, $6 \times 5 \times 4 = 120$, tells you exactly how many different first-second-third outcomes are possible — and thus how many plaques you'd need engraved.

What if the plaque includes all 6 places? Using the same reasoning, you'd fill in six blanks instead of three: $6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$ plaques engraved. We use this process often enough that we gave it a name. We say "I need 6 factorial plaques engraved." When we write a *factorial*, we use an exclamation point:

$$6! = 6 \times 5 \times 4 \times 3 \times 2 \times 1 = 720$$

We use the word "permutation" to mean a particular ordering. This rule says n items can be ordered in $n!$ ways. Thus, mathematicians actually say "If you have a list of n items then we can generate $n!$ different permutations of those items". We will discuss this further in the next chapter.

In Python, there is a `factorial` function in the math library:

```
> python3
>>> import math
>>> math.factorial(6)
720
```

Handy, right? Now you don't need to write a loop to calculate factorials.

Remember when we only wanted the first three names on the plaque? We can solve that problem using factorials:

$$6 \times 5 \times 4 = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = \frac{6!}{3!}$$

This formulation makes it easy to figure out on any calculator with a "!" button.

The rule on this is to fill m positions from n items. It can be done this many ways:

Permutations

For permutations where order is a factor, the formula for ${}^{n}P_{m}$ (n objects pick m) is

$$\frac{n!}{(n-m)!}$$

1.1 Choose

Let's say that there are 12 kids in a classroom and you need a team of four to clean the top of the desks. How many different possible teams are there? You know that if you were giving out four different positions (Like the race gave out 1st, 2nd, and 3rd), the answer would be $12 \times 11 \times 10 \times 9$ or $12!/(12-4)!$.

However, once we pick the 4 people, we don't care what order they are in, right? In this problem, the team "Anne, Brad, Carl, and Don" is the same as the team "Carl, Don, Brad, and Anne".

Thus, the quantity $12!/(12-4)!$ is many times too large, because it counts each permutation separately. To get the right number, we just divide this by the number of possible permutations for a group of four people: $4!$

That gets us our answer: How many different teams of four can be chosen from 12 people?

$$\frac{12!}{(12-4)!4!} = 495$$

Generally, this formula, referred to as the *Binomial coefficient*. In combinatorics, we use this quantity a lot, so we have given it a name: *choose*

Combinations

For combinations, where order doesn't matter, the formula for ${}^{n}C_{k}$ (n objects choose k) is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

We have also given it a notation. "12 choose 4" is written like this:

$$\binom{12}{4}$$

Python has the `math.comb` function:

```
> python3
>>> import math
>>> comb(12, 4)
495
```

We briefly saw this in the common polynomial products chapter. The binomial theorem says that the coefficients of the terms in the expansion of $(a+b)^n$ are given by the binomial coefficients $\binom{n}{k}$.

$$(a+b)^4 = 1a^4+4a^3b+6a^2b^2+4ab^3+1b^4 = \binom{4}{0}a^4+\binom{4}{1}a^3b+\binom{4}{2}a^2b^2+\binom{4}{3}ab^3+\binom{4}{4}b^4$$

1.2 Fractional Binomial Theorem

An issue arises when we have fractional exponents for this theorem. For example, what is $\sqrt{1+5x} = (1+5x)^{1/2}$? We can use the binomial theorem to expand this as follows:

$$(1+5x)^{1/2} = \binom{1/2}{0}1 + \binom{1/2}{1}5x + \binom{1/2}{2}(5x)^2 + \binom{1/2}{3}(5x)^3 + \ldots$$

Pascal's triangle only works when the exponent n is a non-negative integer. Here, the exponent is $\frac{1}{2}$, not a whole number. There's no ”row $\frac{1}{2}$” of Pascal's triangle, so we need to extend the pattern.

This extension is called the ***general binomial theorem***. To make sense of $\binom{1/2}{k}$, we need a definition of the binomial coefficient that doesn't rely on factorials, since $(1/2)!$ doesn't make sense. Recall that

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n(n-1)(n-2)\cdots(n-k-1)}{k!}$$

Notice that the right-hand side only ever multiplies and divides — it never actually requires n to be a whole number! This gives us a definition that works for *any* real number n, including fractions and negative numbers:

Generalized Binomial Coefficient

For any real number n and non-negative integer k,

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}$$

When n is *not* a non-negative integer, like $n = \frac{1}{2}$, none of the factors $n, n-1, n-2, \ldots$ is ever zero, so the expansion never terminates. Instead of a finite row of Pascal's triangle, we get an infinite series.

General Binomial Theorem

Recall that for $|x| < 1$, the geometric series gives

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots$$

Since $\frac{1}{1-x} = (1-x)^{-1}$, this looks like a binomial expansion where the exponent -1 is not a positive integer. Using the generalized binomial coefficient,

$$\binom{n}{k} = \frac{n(n-1)(n-2)\cdots(n-k+1)}{k!}$$

For any real number n and $|x| < 1$,

$$(1+x)^n = \sum_{k=0}^{\infty} \binom{n}{k} x^k = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

this is called the *binomial series*, and it generalizes the binomial theorem to arbitrary real exponents. Setting $n = -1$ recovers the geometric series above.

The condition $|x| < 1$ matters here for convergence. When n was a positive integer, the expansion was just a polynomial, and polynomials are perfectly well-behaved for every value of x. But now that we have an infinite series, we need the terms to actually get smaller and smaller so the sum settles down to a finite value. If $|x| \geq 1$, the terms don't shrink, and the series doesn't converge to anything useful.

Let's compute the coefficients for $\binom{1/2}{k}$

$$\begin{aligned}
\binom{1/2}{0} &= 1 \\
\binom{1/2}{1} &= \frac{1}{2} \\
\binom{1/2}{2} &= \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!} = \frac{\frac{1}{2}\cdot\left(-\frac{1}{2}\right)}{2} = -\frac{1}{8} \\
\binom{1/2}{3} &= \frac{\frac{1}{2}\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!} = \frac{3/8}{6} = \frac{1}{16}
\end{aligned}$$

So,

$$(1+x)^{1/2} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 - \cdots$$

Now we can go back to our original problem, $\sqrt{1+5x} = (1+5x)^{1/2}$. This is just $(1+x)^{1/2}$ with x replaced by $5x$, so we substitute $5x$ everywhere:

$$\begin{aligned}
\sqrt{1+5x} &= 1 + \frac{1}{2}(5x) - \frac{1}{8}(5x)^2 + \frac{1}{16}(5x)^3 - \cdots \\
&= 1 + \frac{5}{2}x - \frac{25}{8}x^2 + \frac{125}{16}x^3 - \cdots
\end{aligned}$$

This expansion is only valid for $|5x| < 1$, i.e. $|x| < \frac{1}{5}$.

Exercise 1 Fractional Binomial Expansion

Working Space

1. Find the first four terms in the expansion of $(1+x)^{-1}$, in ascending powers of x. For what values of x is this expansion valid?
2. Find the first three terms in the expansion of $\sqrt{1-2x}$, in ascending powers of x.

Answer on Page 9

1.3 Applications in Probability

Suppose you flip a fair coin 5 times. What's the probability that you get exactly 3 heads?

Note that there are only two options: Heads (H) or Tails (T). Thus, there are $2^5 = 32$ possible sequences of flips. Each sequence is equally likely, so the probability of any particular sequence is $\frac{1}{32}$. Only a certain number of sequences will have exactly 3 heads.

One way to get 3 heads is HHHTT. There's also HTHHT, THHHT, and many other orderings also give you 3 heads. Before we can find the probability, we need to know *how many* different sequences of 5 coin flips contain exactly 3 heads. As we have discussed, we can use the n-choose-k formula. There are

$$\binom{5}{3} = \frac{5!}{3!2!} = 10$$

such sequences. This is said "5 choose 3", or, more explicitly "get 3 successes in 5 trials".

Each individual sequence, like HHHTT, has probability $\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2$ of occurring, since each flip is independent. Since there are 10 equally likely sequences that give exactly 3 heads, we add up 10 copies of that probability:

$$P(\text{exactly 3 heads}) = \binom{5}{3}\left(\frac{1}{2}\right)^3\left(\frac{1}{2}\right)^2 = 10 \cdot \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$$

This pattern of counting the arrangements with $\binom{n}{k}$, then multiplying by the probability of one arrangement works whenever you have repeated *independent* trials with only two possible outcomes each time (success or failure).

Exercise 2 Free Throws

A free throw shooter has a past record of making 60% of her free throws. Over the next 6 free throws, what is the probability that she makes exactly 4 of them?

Working Space

Answer on Page 9

This probability situation has been rightfully called the *binomial distribution*. This is a brief introduction to the binomial distribution, and we will discuss it in more detail in a later chapter.

This is a draft chapter from the Kontinua Project. Please see our website (`https://kontinua.org/`) for more details.

Appendix A

Answers to Exercises

Answer to Exercise 1 (on page 6)

1. $(1+x)^{-1} = 1 - x + x^2 - x^3 + \cdots$, valid for $|x| < 1$
2. $\sqrt{1-2x} = 1 - x - \frac{1}{2}x^2 - \cdots$

Answer to Exercise 2 (on page 7)

$$P(\text{exactly 4 makes}) = \binom{6}{4}(0.6)^4(0.4)^2 = 0.31104$$

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